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REVIEW ARTICLE

Risk prediction in spine surgery: a scoping review of traditional models, artificial intelligence, and the challenge of clinical translation

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Abstract

Background Accurate perioperative risk stratification is essential to patient safety and informed consent in spine surgery. Traditional regression-based risk scores are widely used but have modest predictive performance, while artificial intelligence and machine learning (AI/ML) approaches are increasingly applied to surgical risk prediction. This review synthesizes evidence on spine surgery risk prediction models, compares traditional approaches with AI/ML methods, and identifies key translational barriers.

Methods A structured literature search was conducted across major biomedical databases using combinations of spine surgery, risk prediction, perioperative outcomes, and AI-related terms, with additional conceptual searches targeting explainability, validation, and clinical translation. Studies were selected for relevance to adult spine surgery risk prediction, model development or validation, and methodological or translational considerations. Given substantial heterogeneity in study design and outcomes, findings were synthesized qualitatively using a narrative approach.

Results Traditional spine-specific risk models demonstrate fair-to-good discrimination for common outcomes, with typical areas under the receiver operating characteristic curve (AUCs) ranging from approximately 0.64 to 0.78. AI/ML models often report modest improvements in discrimination over regression-based approaches, particularly for common outcomes such as intensive care unit (ICU) admission and mortality, but gains are inconsistent and context-dependent. Across both model types, external validation, calibration drift, limited prospective outcome evidence, and challenges related to interpretability and workflow integration remain prominent.

Conclusions Traditional risk models remain interpretable, trusted, and competitively performant for many spine surgery outcomes. While AI/ML approaches expand data integration and interaction modeling, their clinical impact is constrained by validation, trust, and implementation barriers. Future progress will depend less on incremental performance gains and more on rigorous external validation, prospective outcome studies, and integration into clinical workflows.

Keywords Spine surgery · Risk prediction · Machine learning · Clinical decision support · Model validation · Clinical translation

Introduction

Risk stratification is a cornerstone of patient safety, informed consent, and perioperative resource allocation in spine surgery, where the spectrum of operative complexity and patient comorbidity results in substantial heterogeneity in postoperative outcomes [1, 2]. Accurate preoperative risk estimation enables surgeons to engage in meaningful shared decision-making, align patient expectations with realistic outcomes, and tailor perioperative planning to individual risk profiles. From determining the need for preoperative optimization to anticipating intensive care unit (ICU) utilization and discharge disposition, risk prediction directly

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informs both clinical and operational decision-making in modern spine practice.

To address these needs, multiple spine-specific risk prediction tools have been developed, including the Universal Spine Score, SpineSage, and RAT-Spine. These models incorporate combinations of demographic, comorbidity, and procedural variables and generally demonstrate fair-to-good discrimination, with reported areas under the receiver operating characteristic curve (AUCs) ranging from approximately 0.64 to 0.78 for common postoperative adverse events [1, 2]. While these tools represent meaningful advances over generic surgical calculators, their predictive performance remains constrained, particularly for complex, multi-level procedures and less frequent but clinically consequential complications such as major medical events, reoperation, or mortality. Furthermore, many of these models were derived from specific institutional or registry datasets, raising concerns regarding generalizability across practice settings and patient populations.

Despite the proliferation of increasingly sophisticated risk models, traditional clinical measures continue to demonstrate robust predictive value. Notably, the American Society of Anesthesiologists (ASA) physical status classification remains strongly associated with perioperative morbidity in spine surgery, with reported odds ratios of approximately 3 for ASA class III patients and 5–6 for ASA class IV patients [3–5]. The persistence of this association across diverse cohorts underscores the enduring utility of simple, global assessments of physiologic reserve. This observation raises an important question as to whether added model complexity consistently translates into clinically meaningful improvements in risk discrimination or calibration.

At the same time, widely used generic surgical risk calculators frequently fail to capture clinically relevant sources of heterogeneity that are particularly salient in spine surgery. Factors such as frailty, sagittal and coronal deformity, prior instrumentation, and psychosocial determinants of recovery are often incompletely represented, or entirely absent, in structured datasets used for model development [4, 5]. As a result, these tools may perform adequately in average-risk populations while systematically miscalibrating risk at the extremes of patient complexity, precisely where accurate prediction is most needed to guide decision-making. In particular, adult spinal deformity corrections involving long-segment constructs, pelvic fixation, and three-column osteotomies carry substantially higher complication rates than single-level decompressions, yet most available risk models do not differentiate among these procedural categories in a granular manner.

Against this backdrop, a critical comparison of traditional rule-based and regression-derived models with newer artificial intelligence (AI) and machine learning (ML) approaches is warranted. While AI-enabled models offer the potential to capture nonlinear interactions and high-dimensional relationships beyond the scope of conventional methods, they also introduce tradeoffs related to interpretability, clinical usability, and implementation burden. Central to this evolving landscape is the question of whether increased algorithmic complexity yields meaningful clinical benefit or primarily produces incremental, context-dependent gains in discrimination at the expense of transparency and trust. The evolution of spine surgery risk modeling, from general surgical indices to spine-specific tools and emerging AI-based approaches, is summarized in Fig. 1.

Methods

A structured literature search was conducted following PRISMA-ScR (Preferred Reporting Items for Systematic Reviews and Meta-Analyses extension for Scoping Reviews) guidelines [6] using combinations of spine surgery, risk prediction, perioperative outcomes, and artificial intelligence-related terms, with additional conceptual searches targeting explainability, validation, and clinical translation of predictive models. Searches were performed across PubMed, Embase, Scopus, and Web of Science from database inception through December 15, 2025. Reference lists of included studies were also reviewed to identify additional relevant publications. The study identification and selection process are summarized in a PRISMA-ScR flow diagram in Fig. 2.

Titles and abstracts were screened to identify studies relevant to adult spine surgery risk prediction. Full-text review included studies that developed or validated perioperative risk models in spine surgery, applied artificial intelligence or machine learning methods relevant to surgical risk prediction, or addressed methodological considerations related to interpretability, validation, or clinical adoption. Studies were excluded if they focused solely on non-clinical engineering applications, image segmentation without outcome modeling, or editorial and commentary content.

Substantial heterogeneity across study design, data source, outcome definition, and modeling approach precluded meta-analysis. We used PROBAST (Prediction model Risk Of Bias Assessment Tool) to assess risk of bias in all primary prediction model development and validation studies across four domains: participants, predictors, outcome, and analysis [7]. Systematic reviews, methodology papers, and regulatory or ethical analyses ($n = 22$) were classified separately because PROBAST does not apply to these study types. Domain-level and overall risk-of-bias ratings are provided in Supplementary Table S1.

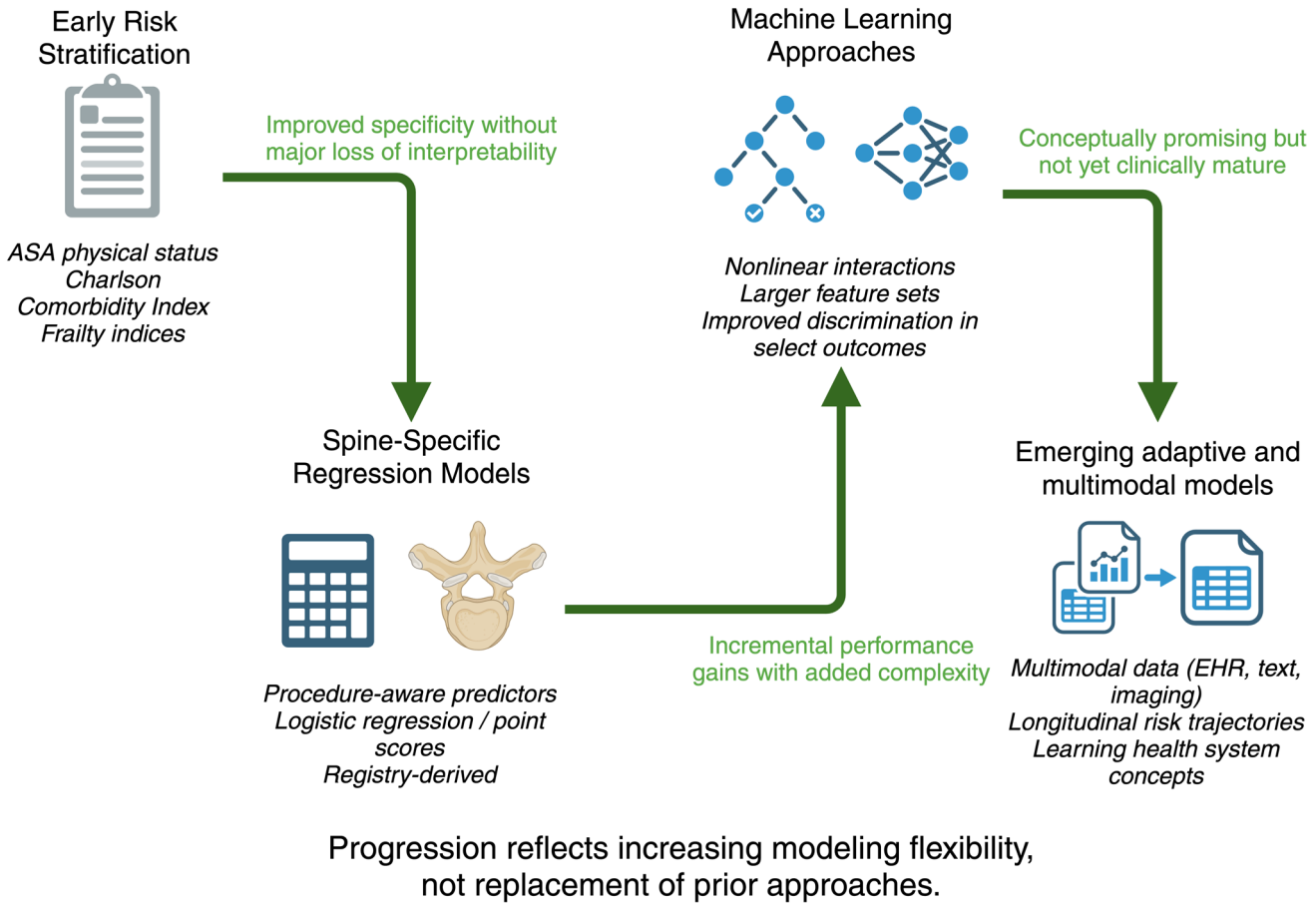


Fig. 1 *Evolution of spine surgery risk modeling approaches.* Conceptual timeline illustrating progression from general surgical risk indices to spine-specific regression models, machine learning-based approaches, and emerging adaptive or multimodal concepts. The figure emphasizes increasing modeling complexity and data integration while highlighting continuity rather than replacement across approaches. *Created in BioRender. Kumar, R. (2025)* https://BioRender.com/8dvm0w0

Results

The literature search identified 19,050 records (18,938 from database searching and 112 from additional sources including reference lists and citation tracking). After removing duplicates, 14,620 records remained for title and abstract screening. Following screening, 403 full-text articles were assessed for eligibility, of which 362 were excluded. Forty-four studies met inclusion criteria and were included in the narrative synthesis (Fig. 2). Included studies spanned several thematic areas: traditional spine-specific risk models and validation studies ($n = 12$), AI/ML model development and comparison studies ($n = 17$), methodological and validation studies ($n = 8$), and regulatory, ethical, and implementation considerations ($n = 7$). Common terminology across the literature included "risk calculator," "prediction model," and "clinical decision support," though definitional consistency varied substantially across studies. The discussion that follows is organized into five thematic subsections: (1) foundations and traditional spine risk models, (2) emergence of AI and machine learning approaches, (3) advanced AI approaches, interpretability, and trust, (4) validation, generalizability, and real-world performance, and (5) regulatory, ethical, and translational barriers.

PROBAST identified substantial methodological limitations across the 23 primary prediction model development and validation studies (Supplementary Table S1). Twelve studies (52.2%) had high overall risk of bias, most commonly because they relied on single-center retrospective cohorts and had important analytic limitations. Seven studies (30.4%) had unclear risk of bias, whereas only four (17.4%) had low risk of bias across all four PROBAST domains. Among spine-specific prediction model studies, only 1 of 13 (7.7%) achieved low overall risk of bias. The remaining 22 references, including systematic reviews, methodology papers, and regulatory or ethical analyses, were not eligible for PROBAST assessment.

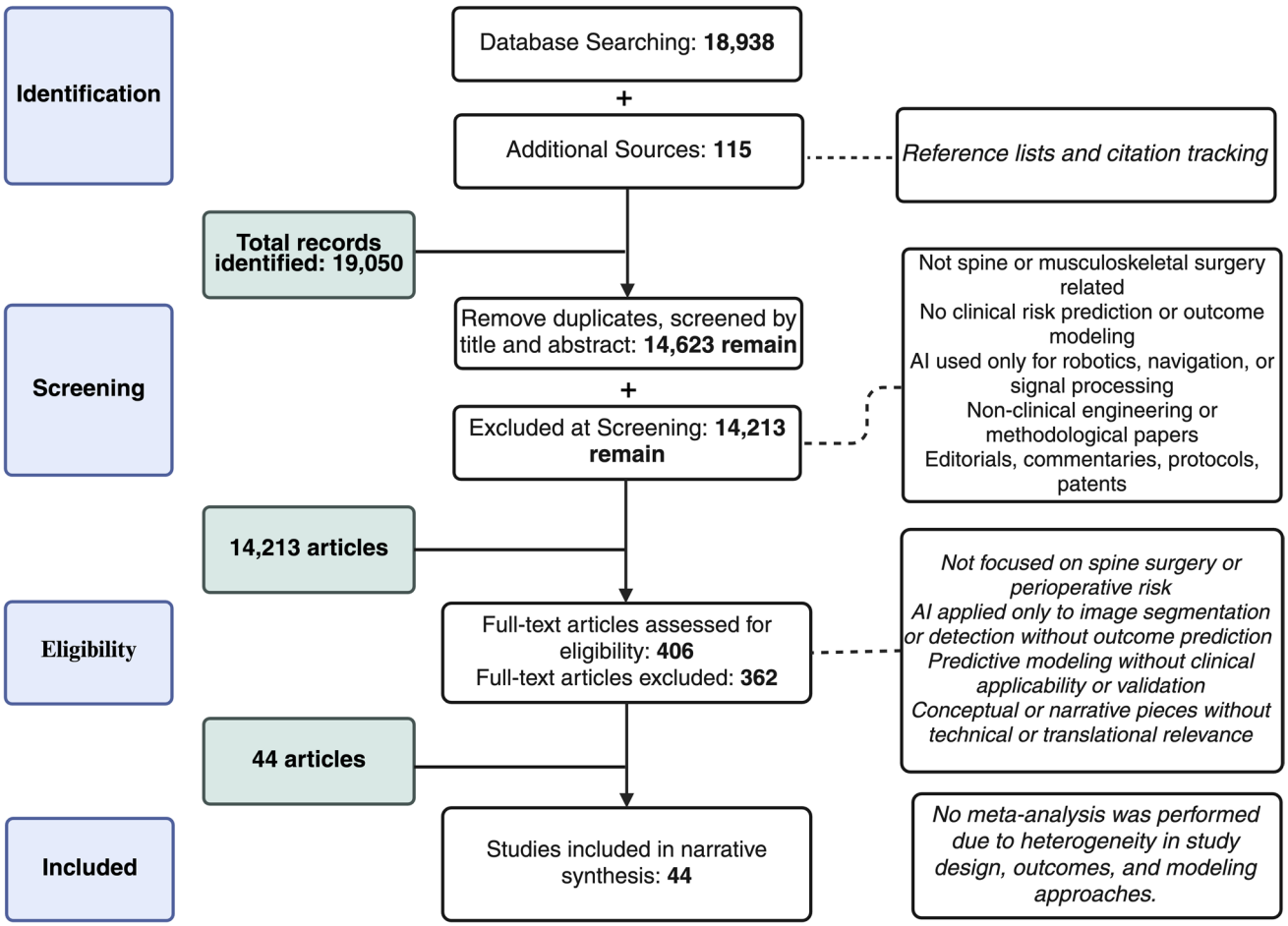


Fig. 2 *Modified PRISMA-ScR flow diagram for literature identification in a scoping review*. This figure summarizes the structured identification, screening, eligibility assessment, and inclusion of studies informing this scoping review on adult spine surgery risk prediction models and their validation and translational considerations, spanning traditional regression-based tools and AI/ML approaches. The flow diagram is provided for transparency of literature selection and does not indicate that this work was conducted as a systematic review or meta-analysis. *Created in BioRender. Kumar, R. (2025)* https://BioRender.com/27nf7dm

Discussion

Foundations and traditional spine risk models

Early spine surgery risk stratification was largely adapted from general surgical indices such as ASA class, the Charlson Comorbidity Index, and early frailty measures [3–5]. These tools offered a pragmatic foundation but were not designed to address spine-specific procedural complexity or neurologic risk. Representative traditional spine surgery risk models, their primary outcomes, typical performance characteristics, and principal limitations are summarized in Table 1.

Subsequent models incorporated spine-relevant predictors, including surgical invasiveness, number of levels, approach, and revision status, reflecting recognition that outcomes are driven by interacting patient, anatomic, and technical factors [1, 2, 8]. Traditional spine risk models have therefore been dominated by point-based scores and logistic regression calculators derived from registry or institutional datasets.

The strengths of these models include simplicity, transparency, low data burden, and strong clinician trust, facilitating bedside use and integration into preoperative discussions. Across studies, discrimination typically falls within the fair-to-good range, with AUCs of approximately 0.64 to 0.78 [1–3], and in many clinical contexts, these tools perform comparably to more complex approaches.

However, limitations arise from static, additive assumptions that constrain the modeling of nonlinear relationships and interactions among risk factors [15, 16]. External validation frequently reveals calibration drift across institutions and populations, raising uncertainty regarding which traditional models remain adequate for routine care and which warrant replacement [2, 8]. External validation studies of spine-specific models demonstrate substantial performance

Table 1 Representative traditional spine surgery risk models and indices

Model name	Primary outcome(s) predicted	Key input domains	Typical discrimination (AUC / c-statistic)	Strengths and limitations
Broda et al. elective cervical/lumbar model [1]	Any perioperative complication; unplanned ICU admission in elective cervical and lumbar surgery	Demographics; comorbidities (including ASA); BMI; smoking; operative time; number of levels; surgical approach	~0.70–0.78 for composite complications and ICU admission	Spine-specific and procedure-focused using routinely available variables; however, derived from a single center with limited external validation and limited capture of frailty or psychosocial risk [4, 5]
Zehnder et al. complication model (European Spine Journal score) [3]	In-hospital or 30-day perioperative medical and surgical complications	Age; sex; ASA; comorbidities; surgical invasiveness; indication; revision status	~0.70–0.75 for any complication	Multicenter design incorporating global health status and procedural complexity; external validation is limited and calibration may vary across institutions, with no explicit modeling of nonlinear interactions [2, 8]
SpineSage risk calculator [2, 9–11]	Medical and surgical complications, mortality, and resource utilization after spine surgery	Demographics; comorbidities/ASA; BMI; smoking; surgical region; number of levels; approach; operative time	Development AUC ~0.70–0.80; external validation AUC ~0.64–0.72	Widely accessible and spine-specific with reasonable discrimination for common complications; external validation demonstrates reduced accuracy and miscalibration, and limited representation of frailty, psychosocial risk, and detailed anatomic features [2, 4, 5, 8, 13, 14]
NSQIP-based spine complication models (including indices evaluated by Ondeck et al.) [4]	30-day medical and surgical complications, readmission, reoperation	Demographics; comorbidities; ASA; functional status; basic procedural variables	Typically ~0.65–0.75 for composite complications and readmission	Large, standardized registry enables benchmarking; however, models are not spine-specific and lack granularity for anatomic complexity and neurologic status, with calibration degradation in external cohorts [2, 8]
ASA- and comorbidity-based indices (ASA, mCCI, mFI) [4, 5]	Any adverse event, major complication, or mortality after spine surgery	ASA class; age; BMI; comorbidity burden; frailty components	~0.60–0.70; stepwise increase in risk with higher ASA (OR ~3 for ASA III; ~5–6 for ASA IV)	Extremely simple and universally available with consistent signal across procedures; limited ability to capture procedure-specific risk, anatomy, or psychosocial context, with plateauing incremental value beyond age and BMI

Table 1 (continued)

Model name	Primary outcome(s) predicted	Key input domains	Typical discrimination (AUC / c-statistic)	Strengths and limitations
PROM-based lumbar outcome models (Khor, McGirt, Halicka) [9–12]	Pain, disability, and quality-of-life improvement; likelihood of achieving MCID or treatment success	Baseline PROMs (ODI, NRS pain, PROMIS); demographics; comorbidities; procedure type; smoking; symptom duration	~0.64–0.79 for MCID or treatment success	Patient-centered and useful for shared decision-making regarding expected benefit; focused on long-term outcomes rather than perioperative complications, with performance sensitive to baseline symptom severity and incomplete capture of psychosocial and socioeconomic factors

Summary of commonly used spine-specific and general surgical risk tools, including primary outcomes, key input domains, typical discrimination, and principal strengths and limitations. Models are representative rather than exhaustive and illustrate the performance and scope of traditional approaches

degradation when applied to new populations, with SpineSage showing an AUC of 0.75 in external validation compared to higher internal performance, while NSQIP systematically underestimated risk in certain populations [2, 8]. The RAT-Spine model demonstrated good agreement between predicted and observed adverse event rates with an AUC of 0.64 in prospective external validation, though calibration varied across risk strata [8]. Dataset shift resulting from institutional variability in case mix, practice patterns, and temporal changes in surgical techniques contributes significantly to this transportability challenge [17, 18]. A persistent controversy centers on whether marginal gains in discrimination justify increased complexity, reduced interpretability, and added workflow burden when existing tools already provide clinically useful estimates.

These constraints have prompted growing interest in more flexible risk modeling strategies designed to address nonlinearity, interaction effects, and limited transportability. Traditional models demonstrate fair-to-good discrimination but are limited by additive assumptions, restricted interaction modeling, and frequent calibration drift across institutions. External validation is inconsistent, and prospective evidence demonstrating outcome improvement is sparse. Whether incremental complexity beyond simple clinical measures such as ASA class consistently yields clinically meaningful benefit remains uncertain. These findings align with prior reviews showing that machine learning has offered only modest and inconsistent advantages over regression-based models across surgical specialties [16, 19]. The current review extends that literature by focusing specifically on translational barriers, implementation challenges, and the persistent gap between model development and real-world use in spine surgery.

Emergence of AI and machine learning in spine risk prediction

Machine learning approaches have been increasingly applied to perioperative risk prediction, enabling capture of nonlinear relationships through tree-based methods, ensemble learners, support vector machines, and neural networks [15, 16, 20, 21]. Key conceptual differences between traditional regression-based models and AI/ML approaches for spine surgery risk prediction are summarized in Table 2. These models typically leverage large feature sets without prespecified importance, allowing automated interaction discovery but increasing susceptibility to overfitting and spurious associations.

Across perioperative medicine, ML models often exceed traditional statistical approaches by approximately 0.05 to 0.10 AUROC on average [16, 20, 21], although gains are inconsistent. Performance improvements are highly

Table 2 Conceptual comparison of traditional regression-based models and AI/ML approaches for spine surgery risk prediction

Dimension	Traditional models (scores, logistic regression)	AI/ML models (trees, ensembles, SVMs, neural networks)
Interpretability	Generally transparent, with coefficients or point scores directly mapping to clinical variables, supporting face validity and bedside explanation of risk estimates [1–5, 8]	Often opaque or only partially explainable; post hoc methods such as feature importance or SHAP can increase transparency but do not ensure true interpretability or clinical validity [20–25]
Data requirements	Rely on relatively small, predefined variable sets including demographics, comorbidities, ASA class, and basic procedural descriptors that are widely available in registries and EHRs [1–5, 8]	Favor larger feature spaces and higher-dimensional inputs such as full EHR data, time series, imaging, or clinical text, increasing demands on data quality, completeness, and curation [13–16, 20, 21]
Ability to model interactions	Primarily assume additive effects with limited prespecified interaction terms, constraining representation of complex nonlinear relationships among risk factors [1–3, 8, 15, 16]	Natively capture nonlinearities and high-order interactions without prespecification, enabling more flexible modeling of complex patient-procedure relationships [15–18, 20, 21]
Performance gains (typical)	Achieve fair-to-good discrimination for many spine outcomes (AUC or c-statistic approximately 0.64–0.78), often comparable to more complex approaches in well-defined cohorts [1–3, 8–12]	Frequently report modest average AUROC improvements of approximately 0.05–0.10 over regression models in perioperative settings, though gains are outcome-specific and context-dependent [15, 16, 20, 21]
External validation burden	Easier to externally validate and recalibrate using established statistical approaches; several spine models demonstrate documented calibration drift across cohorts [2, 3, 8–12]	External validation is less common and technically more demanding; performance may degrade substantially with dataset shift, and recalibration or updating is more complex [13, 14, 16, 19–21]
Workflow integration	Relatively straightforward to integrate into consent clinics or preoperative workflows as bedside scores, static calculators, or EHR tools with minimal infrastructure [1–3, 8–12]	Depend on robust data pipelines, computational resources, and monitoring infrastructure; real-time use requires advanced EHR interoperability and informatics support [13, 14, 19–21]
Regulatory complexity	Typically considered "locked" statistical tools, with regulatory focus on appropriate validation and labeling rather than continuous modification, aligning with existing paradigms [24, 25]	Pose greater regulatory challenges, particularly for adaptive or continuously learning systems, including change control, real-world performance monitoring, and accountability for updates [24, 25]

Key differences across interpretability, data requirements, performance gains, validation burden, workflow integration, and regulatory considerations are summarized to highlight tradeoffs rather than categorical superiority

outcome- and context-dependent, with limited advantage over well-calibrated regression models in some settings [21].

Recent spine-specific ML studies show a similar pattern. Investigators have reported moderate to good discrimination for models predicting cerebrospinal fluid leakage (XGBoost with SHAP interpretation; AUC 0.73) [26], postoperative urinary retention (combined regression and neural network approaches; AUC 0.75) [27], and prolonged operative time after lumbar fusion (multicenter random forest model; AUC 0.83) [28]. These findings support our broader observation that ML performance in spine surgery is often promising, but usually incremental rather than dramatic.

ML models tend to perform best for common outcomes such as ICU admission, mortality, and resource utilization rather than rare complications [15, 16, 19, 21]. For low-incidence events, AUROC may overstate clinical utility, as high discrimination can coexist with poor precision. Precision-recall metrics such as AUPRC are more informative in this setting but remain infrequently reported [15]. When datasets are imbalanced, AUROC can be misleading as baseline models may achieve high discrimination by predicting predominantly negative outcomes [15]. As a result, it remains unclear which spine surgery outcomes meaningfully benefit from ML-based modeling given tradeoffs in interpretability, implementation complexity, and marginal performance gains. The tradeoff between increasing model complexity and real-world clinical utility is conceptualized in Fig. 3.

These mixed performance patterns highlight the need to move beyond aggregate discrimination metrics toward evaluation frameworks that better reflect clinical decision-making. AI/ML models report modest average improvements (approximately 0.05–0.10 AUROC) over traditional approaches, but gains are inconsistent and highly context-dependent. Performance advantages are most apparent for common outcomes such as ICU admission and mortality rather than rare complications. AUROC may overstate clinical utility in imbalanced datasets, and precision-recall metrics are infrequently reported. As a result, it remains unclear which spine surgery outcomes meaningfully benefit from ML-based modeling.

Advanced AI approaches, interpretability, and trust

More advanced AI approaches extend beyond structured data to unstructured sources such as imaging and clinical text. Deep learning enables automatic feature extraction beyond predefined variables, while natural language processing has shown promise in identifying neurologic symptoms and psychosocial context embedded in clinical notes [13, 14, 20]. However, the incremental clinical value of these features remains uncertain.

Multimodal models integrating EHR data with text or imaging demonstrate technical feasibility but lack robust evidence of improved outcomes or decision-making utility [13, 14, 20]. Temporal modeling introduces the concept of longitudinal risk trajectories rather than static snapshots, but these approaches are data-intensive and remain underexplored in spine-specific applications.

As model complexity increases, issues of interpretability and trust become more prominent. Explainability tools such as feature importance measures, SHAP values, and saliency maps are increasingly used to contextualize

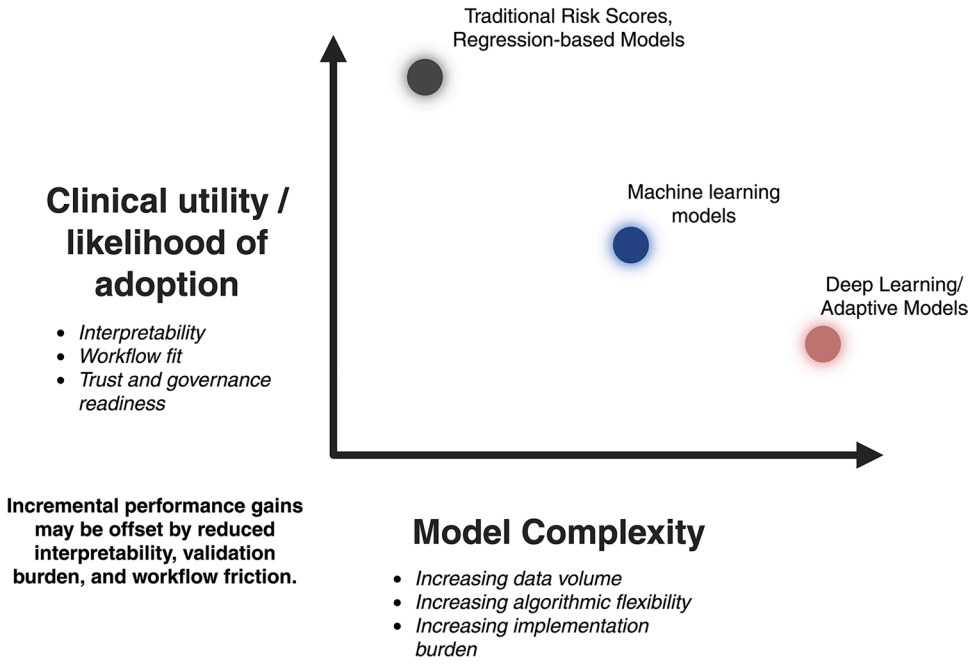


Fig. 3 *Tradeoff between model complexity and clinical utility in spine surgery risk prediction.* Conceptual illustration showing how increasing model complexity, from traditional regression-based scores to machine learning and deep learning approaches, may yield incremental performance gains while introducing challenges related to interpretability, validation, workflow integration, and adoption. The figure emphasizes relative positioning rather than quantitative comparison. *Created in BioRender. Kumar, R. (2025)* https://BioRender.com/foo9f6x

AI predictions [22–24]. SHAP (SHapley Additive exPlanations) values provide model-agnostic explanations by quantifying each feature's contribution to individual predictions, with positive values indicating increased risk and negative values indicating decreased risk [29–32]. In perioperative risk prediction, SHAP analysis has successfully identified clinically meaningful predictors such as age, preoperative red blood cell orders, and C-reactive protein levels as top contributors to mortality risk [31, 33]. While pairing risk estimates with explanations may support clinician sensemaking, explainability does not guarantee interpretability or clinical validity [17, 20, 21]. Studies comparing explainability methods across different ML approaches have demonstrated consistent identification of important clinical features such as age and disease severity, suggesting compatibility with domain knowledge [34]. Overreliance on explanations risks false reassurance, particularly when uncertainty is obscured, or model behavior is oversimplified [22, 24].

Bias and fairness concerns further complicate adoption, with evidence of degraded performance in underrepresented populations and rare conditions, and no consensus standards for acceptable disparities [35–40]. Algorithmic bias can arise from biased training data that reflect historical inequities in healthcare delivery, device-related biases such as pulse oximetry inaccuracies in patients with dark skin, and systemic biases embedded in clinical workflows [36, 39]. No published studies of AI-enabled surgical decision support have reported performance across sociodemographic categories, representing a critical gap in equity assessment [37]. Strategies to address bias include data augmentation to increase demographic diversity, adversarial learning to reduce dependence on protected attributes, and federated learning to combine data from diverse institutions [36, 38]. Ultimately, predictive accuracy alone is insufficient. Risk outputs must be actionable and aligned with clinical decision-making rather than serving solely as numerical estimates.

These challenges highlight the need to evaluate advanced AI systems not only on predictive performance, but also on interpretability, equity, and their ability to support real clinical decisions. Explainability tools such as SHAP and feature importance measures increase transparency but do not ensure clinical validity or true interpretability, and overreliance on explanations may provide false reassurance. Bias and fairness concerns are prominent, with no published studies reporting AI-enabled surgical decision support performance across sociodemographic categories. Standardized fairness metrics and reporting practices remain underdeveloped.

Validation, generalizability, and real-world performance

Internal validation often overestimates model performance, whereas external validation commonly shows substantial decline across institutions [2, 8, 19]. SpineSage and NSQIP-based calculators illustrate this limitation in spine surgery [2], likely because differences in case mix, practice patterns, coding, and outcome measurement reduce transportability. Table 3 summarizes the validation and implementation gaps that affect both traditional and AI-based spine risk models. Although strategies such as recalibration, multicenter training, and periodic updating may improve generalizability, these approaches remain poorly standardized. Just as important, few prospective studies have shown that using these models improves patient outcomes [13, 14, 19]. The MySurgeryRisk platform reflects this gap: it maintained prospective discrimination but did not clearly improve outcomes or produce sustained changes in clinician behavior [13, 14]. Together, these findings suggest that the main barrier to deployment is not predictive performance alone, but whether a model can integrate into clinical workflow, gain clinician trust, and guide decisions in ways that improve care [41, 42]. The high risk of bias across many included studies, especially in the participants and analysis domains, further reinforces the need for multicenter prospective validation before routine clinical use.

Prospective validation studies reveal performance degradation compared to retrospective testing, though well-designed models can maintain clinically useful discrimination [20]. In a large prospective validation of a gradient-boosted ML model for surgical mortality prediction, the AUROC decreased from 0.946 in retrospective testing to 0.956 in prospective deployment, demonstrating stable performance with a sensitivity of 85.3% and a specificity of 91.4% [20]. However, the clinical impact of improved risk prediction on patient outcomes, resource utilization, and decision-making remains largely unproven [21, 43]. A systematic review and meta-analysis of ML-augmented perioperative interventions found that while the Hypotension Prediction Index reduced intraoperative hypotension duration and the Nociception Level Index decreased postoperative pain scores, neither intervention significantly affected hospital length of stay or other downstream clinical outcomes [43].

Workflow integration represents a critical implementation barrier, as AI outputs must be seamlessly incorporated into existing clinical processes rather than adding burden [32, 44, 45]. Successful integration requires treating AI as a routine part of clinical workflow and a member of the care team rather than a substitute for clinical judgment [45]. User interface design, mobile device accessibility, and automated data extraction from electronic health records are essential

Table 3 Validation and translational gaps across spine surgery risk model types

Issue	Traditional models (scores, logistic regression)	AI/ML models (trees, ensembles, neural networks)	Practical implication
External validation	Only a minority of spine-specific risk scores have undergone formal external validation, with frequent performance degradation and miscalibration when applied to new institutions or populations [2, 3, 8]	Even fewer ML models have been externally validated; when evaluated, performance often declines more sharply than for simpler models, reflecting sensitivity to dataset shift and local data pipelines [13–16, 19–21]	Discrimination reported in development cohorts is insufficient for deployment; both model types require systematic external validation, with ML models facing a higher bar for transportability and maintenance
Calibration drift	Calibration drift over time and across institutions is common but often predictable, and recalibration can typically be achieved using relatively simple statistical updates [2, 3, 8]	Calibration drift may be more complex and harder to detect, particularly when models ingest streaming EHR data or undergo implicit updates [16, 20, 21]	Longitudinal calibration monitoring is essential; ML implementations in particular require explicit drift detection and structured update policies rather than one-time deployment
Prospective outcome evidence	Despite technical maturity, prospective evidence demonstrating that traditional spine risk tools improve patient outcomes or meaningfully change clinician behavior remains sparse [2, 3, 8]	Platforms such as MySurgeryRisk demonstrate stable prospective discrimination but show limited or no clear evidence of improved clinical outcomes or sustained behavioral change [13, 14, 19]	Predictive performance alone does not justify implementation; both model classes require prospective, outcome-focused evaluation, with ML models additionally needing to demonstrate that added complexity yields measurable clinical benefit
Bias assessment	Regression-based tools are relatively easier to audit for subgroup performance, but formal bias and fairness analyses are infrequently reported in spine surgery applications [4, 5]	Algorithmic bias is a central concern for ML; emerging studies report unequal performance across demographic and institutional subgroups, while standardized fairness metrics and reporting remain underdeveloped [22–25]	Translation should include structured subgroup and fairness assessments; governance frameworks must explicitly address bias, particularly for higher-stakes ML systems
Governance requirements	Oversight has historically been ad hoc, often limited to local approval or informal review, with tools treated as static calculators after publication [4, 5]	AI/ML tools increasingly trigger formal institutional oversight, with expectations for lifecycle management, monitoring, change control, and clear accountability among developers, clinicians, and health systems [24, 25]	Effective translation requires a shift from a one-time model publication to continuous governance, with explicit institutional policies for approval, monitoring, updating, and decommissioning, especially for ML-based tools

Comparison of common challenges affecting traditional and AI/ML models, with emphasis on external validation, calibration drift, prospective outcome evidence, bias assessment, and governance requirements relevant to real-world clinical deployment

enablers [14, 20]. A mobile application for surgical risk prediction demonstrated the feasibility of delivering automated predictions directly to surgeons' devices with fingerprint login and data loading approximately 10 times faster than web portals [14]. However, human factors challenges persist, including clinician trust, interpretation of model outputs, risk of overreliance or dismissal, and integration with existing cognitive processes [42, 46].

Together, these findings suggest that validation alone is insufficient, and that real-world impact depends on prospective testing, meaningful workflow integration, and demonstrable clinical benefit beyond predictive accuracy. External validation consistently shows performance degradation for both traditional and AI/ML models, while few prospective studies link model use to improved patient outcomes. The MySurgeryRisk platform illustrates stable discrimination without clear outcome improvement. Workflow integration remains a critical barrier, with persistent human factors challenges including clinician trust and risks of over- or under-reliance.

Regulatory, ethical, and translational barriers

Regulatory uncertainty continues to limit the adoption of AI-based software as a medical device, particularly for adaptive systems. Although FDA guidance exists, continuous-learning algorithms challenge traditional approval frameworks [22, 24]. Liability ambiguity regarding clinician versus developer responsibility persists, prompting many institutions to adopt governance committees for model oversight. Workflow barriers, including EHR integration, training burden, and cultural resistance, often outweigh modest performance gains [14, 19, 22]. Best practices for model maintenance, version control, and post-deployment monitoring remain undefined, and debate continues over whether risk tools should estimate risk alone or also recommend clinical actions.

Traditional risk scales already play a routine role in surgical informed consent, with surgeons using model-based complication estimates to guide preoperative counseling and document risk disclosure. AI-based predictions complicate this process by raising new questions about disclosure obligations, liability, and regulatory oversight. Clinicians may face uncertainty about whether they should disclose a high-risk AI estimate during consent and whether reliance on a poorly validated model could increase medicolegal risk. Because clear legal and regulatory standards for AI-based decision support remain lacking, adoption in this setting remains uncertain [41, 42].

Overall, regulatory pathways for adaptive AI systems remain unclear, and liability frameworks are underdeveloped. Workflow barriers frequently outweigh performance gains. Best practices for model governance, version control, and post-deployment monitoring lack standardization. Whether risk tools should provide risk estimates alone or also recommend clinical actions remains an open question.

Conclusion and future directions

Hybrid models combining clinical knowledge with machine learning may improve robustness, interpretability, and clinician acceptance. Dynamic personalization through risk updates reflecting prehabilitation or intraoperative events remains conceptually attractive but underdeveloped. The digital twin concept for spine surgery is aspirational, with no validated implementations or established frameworks for integrating biomechanics, imaging, and longitudinal data.

Prospective evaluation of risk prediction tools in spine surgery might involve randomized trials comparing standard preoperative counseling against model-guided shared decision-making. Ideal trial design would include patient selection criteria (e.g., elective lumbar fusion candidates), clearly defined intervention arms (standard care versus risk model-informed counseling), and primary endpoints such as decision quality, expectation-outcome concordance, patient satisfaction, and perioperative resource utilization. Such trials would need to address challenges including standardization of model outputs across sites, clinician training requirements, and measurement of decision-making processes rather than solely surgical outcomes. Multi-site collaboration would be essential to ensure adequate power and generalizability. In the interim, clinicians can incorporate validated risk estimates into preoperative counseling by presenting model-generated probabilities alongside clinical assessment, framing predictions as one input among many in shared decision-making rather than as definitive forecasts.

In summary, traditional spine risk models remain interpretable, trusted, and competitively performant for many outcomes. AI and machine learning expand data integration and interaction modeling but face persistent barriers related to validation, trust, and workflow integration. The highest-priority needs include rigorous external validation, prospective outcome studies, and implementation science. Incremental performance gains must be weighed carefully against complexity, transparency, and real-world usability before widespread adoption.

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Declarations

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